3. Express the rational number $\frac{1}{33}$ in recurring decimal form by using the recurring decimal expansion of $\frac{1}{11}$. Hence write $\frac{71}{33}$ in recurring decimal form.

4. Express the following decimal expression into rational numbers.

(i) $0.\overline{24}$ (ii) $2.\overline{327}$ (iii) -5.132

(iv) $3.1\overline{7}$ (v) $17.2\overline{15}$ (vi) $-21.213\overline{7}$

5. Without actual division, find which of the following rational numbers have terminating decimal expansion.

(i) $\frac{7}{128}$ (ii) $\frac{21}{15}$ (iii) $4\frac{9}{35}$ (iv) $\frac{219}{2200}$

2.3.6 Decimal Representation to Identify Irrational Numbers

It can be shown that irrational numbers, when expressed as decimal numbers, do not terminate nor do they repeat. For example, the decimal representation of the number π starts with 3.14159265358979..., but no finite number of digits can represent π exactly, nor does it repeat.

Consider the following decimal expansions:

(i) 0.10110011100011111… (ii) 3.012012120121212…

(iii) 12.230223300222333000… (in) $\sqrt{2} = 1.4142135624\ldots$

Are the above numbers terminating (or) recurring and non- terminating? No… They are neither terminating, nor non–terminating and recurring. Hence they are not rational numbers. They cannot be written in the form of $\frac{p}{q}$, where $p, q \in \mathbb{Z}$ and $q \neq 0$. They are irrational numbers.

A number having non- terminating and non- recurring decimal expansion is an irrational number.

Example 2.8 Find the decimal expansion of $\sqrt{3}$.

Solution

$$
\begin{array}{r|l}
 & 1.7320508\ldots \\ \hline
1 & 3.00,00,00,00,00,\ldots \\
 & 1 \\ \hline
27 & 200 \\
 & 189 \\ \hline
343 & \quad 1100 \\
 & \quad 1029 \\ \hline
3462 & \qquad 7100 \\
 & \qquad 6924 \\ \hline
346405 & \qquad\; 1760000 \\
 & \qquad\; 1732025 \\ \hline
34641008 & \qquad\quad 279750000 \\
 & \qquad\quad 277128064 \\ \hline
 & \qquad\qquad 2621936
\end{array}
$$

We often write $\sqrt{2} = 1.414$, $\sqrt{3} = 1.732$, π = 3.14 etc. These are only approximate values and not exact values. In the case of the irrational number π, we take frequently $\frac{22}{7}$ (which gives the value $3.\overline{142857}$) to be its correct value but in reality these are only approximations. This is because, the decimal expansion of an irrational number is non-terminating and non-recurring. None of them gives an exact value!